Chapter 11

Modeling of citric acid production using SuperPro Designer*

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Chapter outline

11.1 Introduction **221**
11.2 Process description **223**
11.2.1 Fermentation section 226
11.2.2 Isolation section 230
11.3 Model setup highlights **231**
11.3.1 Material charges 231
11.3.2 Modeling the fermentation step 233
11.3.3 Modeling the cleaning operations 237
11.4 Scheduling setup **238**
11.4.1 Operating in staggered mode 238
11.4.2 Operating with independent cycling 239
11.4.3 Calculating the minimum cycle time 239
11.5 Process simulation results **242**
11.6 Process scheduling and debottlenecking **242**
11.7 Process economics **245**
11.7.1 Capital investment costs 245
11.7.2 Operating costs 246
11.7.3 Economic evaluation 247
11.8 Variability analysis **248**
11.9 Conclusions **250**
Exercises **251**
Exercise 1: Decreasing the cycle time 251
Exercise 2: Increasing the batch size 251
Acknowledgments **252**
References **252**
Further reading **252**

11.1 Introduction

Processing in batch or semi-continuous mode is an operational approach adopted by industries in many important sectors of economy such as pharmaceuticals, biotech, consumer goods, and food. Even though the traditional

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chemical and petrochemical industries (mainly operating in continuous mode and producing bulk commodity products) are reaping the benefits of process simulation for many decades, batch, and semi-continuous process industries (many of them producing high-value-added products) were slower in adopting this technology. With the domination of a global competitive market, all process industries cannot afford to overlook any possible improvement in their production efficiency, including those originated from process simulation.

SuperPro Designer (www.intelligen.com) is a process simulator developed in the early 1990s to address the specific modeling needs in batch/semi-continuous processing. Even though its initial emphasis was in the biotechnological sector, its scope was expanded to cover integrated batch and continuous processes in a wide range of industries, such as pharmaceutical, specialty chemical, food processing, metallurgical, consumer goods, and environmental protection.[1] Process simulation in these industries can be used for preliminary design, retrofit of existing processes, throughput analysis, estimation of capital and operating costs, as well as to assess the environmental impact of processes.

Process simulation can play an important role in all stages of a product or process development. In biotechnology and pharmaceutical industries, by the time detailed engineering work has been initiated, a process is more than 80% fixed and the majority of important decisions for capital expenditures and product commercialization are already made. Furthermore, stringent regulatory in the biotechnology and pharmaceutical industry restricts process modifications after clinical efficacy of the drug is established. It is therefore suggested that process modeling and evaluation be initiated at the early stages of product development (Carmichael et al., 2018).

In this chapter, SuperPro Designer will be demonstrated for a citric acid production case study. Citric acid is a commodity organic acid used in the food and beverage industries to preserve and enhance flavor (Crueger & Crueger, 1989). Among all fruit acids used in the process industries, citric acid plays an exceptional role with a global annual production that surpassed 2 million tons in year 2018, and is projected to reach a volume of nearly 3 million tons by year 2024 (Cision, 2021). Around half of this production is consumed by the beverages industry. Citric acid is the main organic acid produced today by fermentation (Berovic & Legisa, 2007). In the food industry, citric acid is used extensively in carbonated beverages to provide taste and complement fruit flavors, in jams and jellies for taste and for pH adjustment, in frozen fish and shellfish to prolong their shelf-life. It is also used in various foods to increase the effectiveness of antimicrobial preservatives, as well as to enhance the action of antioxidants and inactive enzymes. Since it is biodegradable, it is readily metabolized and safe for both consumers and industry. It is also used in the pharmaceutical industry for various purposes, e.g., to maintain the stability

1. See Chapter X for the modeling of wastewater treatment plant.

of active ingredients, to enhance the activity of preservatives, pH adjustment in the cosmetics industry, and as metallic-ion chelator in antioxidant systems (Apelblat, 2014; Berovic & Legisa, 2007).

Fermentation is the most commonly used method for bulk production of citric acid. Low-cost and readily available carbohydrates are used as raw materials for its production. Beet and cane molasses, maltose syrups, hydrolyzed starch (corn, wheat, tapioca, and potatoes), cellulose hydrolysates, and waste products of the sugar industry are utilized in different countries depending on their local availability (Apelblat, 2014). High quality molasses is preferred because it reduces the pretreatment costs before it is used in fermentation (Berovic & Legisa, 2007).

Even though different microorganisms have been studied, strains of filamentous fungus *Aspergillus niger* are by far the most commonly used microorganisms. The most important advantages that led to its extensive use were their ability to grow at low pH and the high product yields. *Aspergillus niger* is easy to handle, tolerant to high concentrations of substrates, and can ferment various cheap raw materials. Mutagenesis and strain selection have been carried out to create industrial strains with even better performance (Show et al., 2015).

Citric acid is commonly precipitated as calcium citrate by the addition of lime (calcium hydroxide) and recovered after treatment with sulfuric acid. The disadvantage of this method is that gypsum is produced as an undesired byproduct. Alternative technologies have been proposed to eliminate the formation of gypsum such as solvent extraction, ion-exchange adsorption, membrane filtration, and electrodialysis (Apelblat, 2014). Butanol has been used as an extractant, as has tributyl phosphate (Roberts, 1979). Ion pair extraction by means of secondary or tertiary amines dissolved in a water-immiscible solvent (e.g., octyl alcohol) provides an alternative route. The use of electrodialysis membranes which allows the recovery of citric acid directly from the fermentation broth is also an attractive alternative (Blanch & Clark, 1997).

In this case study, citric acid is produced via fermentation using *Aspergillus niger*. The data used in the development of the model were obtained from the open research literature. The case study is a modified version of an example found in SuperPro Designer software (Intelligen Inc., 2021) with a more detailed depiction of the steps involved in the development of the model and extended analysis of the results.

11.2 Process description

Fig. 11.1 shows the complete process flowsheet developed in SuperPro Designer for the production of citric acid with the use of molasses as the carbon source for fermentation. It is a large-scale batch process utilizing approximately 139.8 metric tons (MT) of molasses per batch and producing 54.8 MT of dehydrated citric acid crystals of high purity (99.5%). The

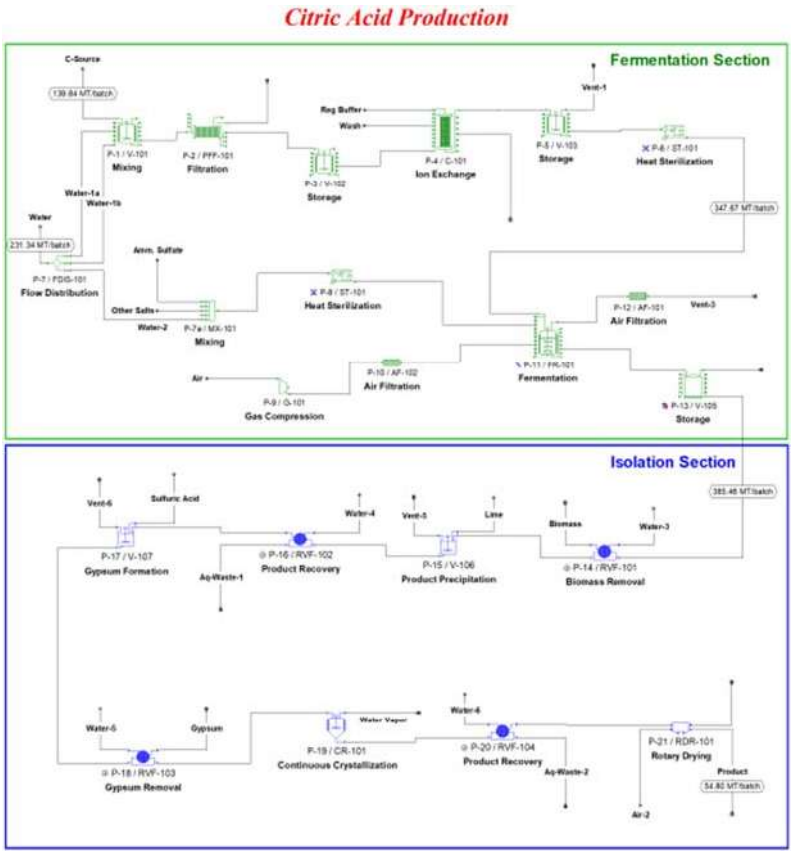

FIGURE 11.1 The citric acid production flowsheet.

corresponding annual amounts are 45,868 MT of molasses and 17,973 MT of citric acid. Table 11.1 shows the assumed (simplified) composition of the stock mixture Molasses. As shown, "Glucose," $C_6H_{12}O_6$, is used to represent all fermentable sugars and "NFS" represents the nonfermentable substances. The component "Impurities" collectively represents all heavy metals and other impurities that must be removed because they are detrimental to the growth of the microorganism. Apart from molasses, three additional stock mixtures are registered in the model representing buffers used in the process, e.g., "H_2SO_4 (10% w/w)," "Lime (33%)," and "NaOH (1 M)"; their compositions are also given in Table 11.1.

The declaration of components in the SuperPro Designer model is completed with the inclusion of ammonium sulfate, $(NH_4)_2SO_4$, used as a nitrogen source in the fermentation, "Nutrients" representing all other salts fed into the fermenter, citric acid ($C_6H_8O_7$) in diluted from, citric acid in crystal form ("CA Crystal") and various reagents used (sodium hydroxide, NaOH;

TABLE 11.1 Composition of stock mixture used in simulation.

Streams	Component	(%w/w)
Molasses	Glucose	50
	NFS	20
	Impurities	1
	Water	29
H_2SO_4 (10% w/w)	H_2SO_4	10
	Water	90
Lime (33%)	$Ca(OH)_2$	33
	Water	67
NaOH (1 M)	NaOH	3.84
	Water	96.16

calcium hydroxide, $Ca(OH)_2$; sulfuric acid, H_2SO_4) or produced (calcium citrate, $C_6H_5O_7Ca_3$; gypsum, $CaSO_4$) in the downstream process. The components CA Crystal, Calcium Citrate, Impurities, NFS, and Nutrients are user-defined components. The component Water was used as a reference component for the initialization of the properties of all user-defined components with the exception of Ca Crystal for which citric acid was used. All other components used in the model and their properties are originated from the SuperPro Designer database.

The process is divided into two sections: "Fermentation" and "Isolation." In SuperPro Designer, a section represents a group of procedures that serve a common processing objective (e.g., the section "Isolation" in this case study encompasses all product purification procedures). Sections can also have different economic parameters. It follows that reporting of all model outputs (yields, resources consumption, etc.) and economic results can be done on a per-section level. In this example, the Fermentation section includes the media preparation, treatment and fermentation procedures. The inoculum preparation steps were ignored for brevity. The Isolation section processes the fermentation broth and includes all purification and isolation procedures that lead to the production of high-purity crystal citric acid. Details for all processes in the "Fermentation" and "Isolation" sections are described below.

Note that all procedures in SuperPro Designer are tagged by their ID, the hosting equipment ID, and a textual description of the procedure. All these three tags are shown below every procedure icon (see Fig. 11.1). In the model formulation of batch processes, it is important to differentiate the processing

steps (comprising the "recipe") from the equipment used; this is because the same equipment could be used to carry out multiple process steps at different times. This is why SuperPro Designer adopts a dual naming approach in each process icon with separate IDs for the procedure and the equipment (SuperPro Designer, User Guide, 2021).

The important input parameters for all operations in the process are shown in Table 11.2. It should be noted that in the context of batch processing, scheduling information (i.e., the duration of operations and how their start/end is defined) is equally important with operating condition parameters. The set-up time for all operations has been set to 0. Also, all input streams are fed in ambient conditions (25°C and 1 atm).

11.2.1 Fermentation section

Molasses is used as carbon source for the fermentation process. It is diluted with water from 50% fermentable sugar content to 20% which is favorable for the growth of *Aspergillus niger*. The process is carried out in mixing tank P-1/V-101, and includes three operations: charging of 100 MT of water into the tank, charging of molasses, and charging of extra water so that the concentration of sugars in the tank drops to 20%. Suspended particulate material is then removed by filtration (P-2/PFF-101) and the media solution is stored in tank P-3/V-102.

Metal ions are subsequently removed by an ion exchange chromatography column (P-4/C-101). The column operates in a cycle consisting of the load, elution, and washing steps. NaOH (1M) is used as the elution buffer and water is used for column washing. The purified solution is collected in a tank (P-5/V-103), heat-sterilized (P-6/ST-101), and fed into the fermenter (P-11/FR-101). In parallel, nutrients (ammonium sulfate and nutrients) are dissolved in water (P-7a/MX-101), heat-sterilized (P-8/ST-101), and fed to the fermenter (P-11/FR-101) after molasses are fed. The same sterilizer (ST-101) is used for the sterilization of both the molasses (P-6) and the nutrients (P-8), but at different times.

Fermentation is carried out according to the following mass stoichiometry:

$$\begin{aligned} &2(\text{Ammonium sulfate}) + 180(\text{Glucose}) + 12(\text{Nutrients}) + 56\,O_2 \rightarrow \\ &16(\text{Biomass}) + 30\,CO_2 + 154(\text{Citric Acid}) + 50\,H_2O \end{aligned} \tag{11.1}$$

The extent of the fermentation reaction is set to 99% and the fermentation is carried out over a period of 5 days at a temperature of 35°C. Air is supplied by a compressor (P-9/G-101) and purified by filtration (P-10/AF-102) at a rate of 0.3 VVM (volume of air per volume of liquid per minute). Cooling water removes the heat produced by the exothermic process and maintains a constant temperature. Once fermentation is completed, the broth is discharged into the holding tank P-13/V-105, which acts as a buffer tank between the upstream

TABLE 11.2 Operating and scheduling data used in the model.

Operation	Operating conditions	Scheduling data
P-1/V-101		
Charge water	100 MT of water	30 min, start with batch start
Charge carbon source	139.8 MT of molasses	60 min, start at end of Charge water
Dilute to 20% (pull-in)	109.4 MT of water	30 min, start at end of Charge carbon source
Transfer to filter	349.2 MT of contents	Duration set by filtration, start at end of Dilute to 20%
P-2/PFF-101		
Filter	Retained: 90% of impurities Cake porosity: 0.3 v/v Filtration flux: 200 L/m^2h	4 h, start with Transfer to filter in P-1
P-3/V-102		
Transfer-in		Duration and start set by Filter in P-2
Transfer-out		Duration set by load in P-4, start at end of Transfer-in
P-4/C-101		
Load	Resin binding capacity: 30 g/L Linear velocity: 1000 cm/h Retained: 100% of impurities	11.5 h, start with Transfer-out in P-3
Elute	2 BV of NaOH (1M) buffer	24 min, starts with end of Load
Wash	2 BV of water	24 min, starts with end of Elute
P-5/V-103		
Transfer-in		Duration and start set by Load in P-4
Transfer-out		Duration set by Sterilize molasses in P-6, start at end of Transfer-in
P-6/ST-101		
Sterilize molasses	Sterilization temp: 140°C Regenerator cold stream out: 60°C Exit temp: 35°C Heat transfer coefficient: 2700 W/m^2K	8 h, start with Transfer-out in P-5

Continued

TABLE 11.2 Operating and scheduling data used in the model.—cont'd

Operation	Operating conditions	Scheduling data
P-8/ST-101		
Sterilize salts	Sterilization temp: 140°C Regenerator cold stream out: 60°C Exit temp: 35°C Heat transfer coefficient: 2700 W/m^{2}K	4 h, start with Pull-in nutrients in P-11
P-11/FR-101		
Transfer-in molasses solution		Duration and start set by Transfer-out in P-6
Pull-in nutrients	Nutrients equal to 8% of molasses	Duration set by Sterilize salts in P-8, start at end of Transfer-in molasses solution
Inoculate		30 min, start at end of Pull-in nutrients
Ferment	Fermentation temperature: 35°C Air flow: 0.3 VVM Conversion: 99% of limiting component Reaction Enthalpy: 2990 kcal/kg of citric acid at 35°C Venting pressure: 1 atm	5 days, start at end of Inoculate
Transfer-out	Rate: 140,000 L/h	2.24 h, start at end of Ferment
CIP	Agent flowrate: 30 L/min·m	Start at end of Transfer-out
	Water flush 1	10 min
	NaOH (1M) flush	30 min
	Water flush 2	10 min
P-13/V-105		
Store		Duration and start set by Transfer-out in P-11

TABLE 11.2 Operating and scheduling data used in the model.—cont'd

Operation	Operating conditions	Scheduling data
P-14/RVF-101 (operating in batches with cycle time of 31 h)		
Apply pre-coat		1 h
Filter	Retained: 100% of biomass LOD: 30% Filtrate flux: 250 L/m^{2}h Water wash: 0.6 vol/cake vol	30 h
P-15/V-106 (operating continuously)		
Neutralize	Neutralization temperature: 50°C Neutralizing agent: $Ca(OH)_2$ 33% w/w Agent excess: 5%	
P-16/RVF-102 (operating in batches with cycle time of 31 h)		
Apply pre-coat		1 h
Filter	Retained: 98% of calcium citrate LOD: 30% Filtrate flux: 175 L/m^{2}h Water wash: 0.6 vol/cake vol	30 h
P-17/V-107 (operating continuously)		
Neutralize	Neutralization temperature: 35°C Neutralizing agent: H_2SO_4 10% w/w Agent excess: 4%	
P-18/RVF-103 (operating in batches with cycle time of 31 h)		
Apply pre-coat		1 h
Filter	Retained: 95% of $CaSO_4$, 1% citric acid LOD: 30% Filtrate flux: 250 L/m^{2}h Water wash: 0.6 vol/cake vol	30 h

Continued

TABLE 11.2 Operating and scheduling data used in the model.—cont'd

Operation	Operating conditions	Scheduling data
P-19/CR-101 (operating continuously)		
Crystallize	Evaporation: 50% of H_2O at 100°C Crystallizing component: Citric acid Crystallization yield: 98%	
P-20/RVF-104 (operating in batches with cycle time of 31 h)		
Apply pre-coat		1 h
Filter	Retained: 98% of crystal citric acid LOD: 30% Filtrate flux: 250 L/m^2h Water wash: 0.6 vol/cake vol	30 h
P-21/RDR-101 (operating continuously)		
Dry	Final LOD: 0.5% Specific evaporation rate: 20 $(kg/h)/m^3$ Product temperature: 110°C	

"Fermentation" section and the downstream "Isolation" section operated in continuous mode. This means that, even though the overall process is still batch, these procedures receive and process material in a continuous way with no breaks in-between batches (see Gantt chart in Fig. 11.16 later in this chapter).

11.2.2 Isolation section

Purification starts with the removal of biomass by a rotary vacuum filter (P-14/RVF-101). The clarified fermentation liquor subsequently flows into an agitated reactor vessel (P-15/V-106) where approximately one part of hydrated lime ("Lime (33%)"), for every three parts of liquor, is gradually added in order to precipitate calcium citrate according to the reaction (in molar stoichiometry) in Eq. (11.2):

$$2(\text{Citric Acid}) + 3\,\text{Ca(OH)}_2 \rightarrow 1(\text{Calcium Citrate}) + 6\,\text{H}_2\text{O} \qquad (11.2)$$

The lime solution must be very low in magnesium content in order to avoid the formation of magnesium citrate which, unlike calcium citrate, is relatively

soluble to water. Calcium citrate is then separated by a second rotary vacuum filter (P-16/RVF-102) and the citrate-free filtrate is sent to a wastewater collection tank. The calcium citrate cake is sent to another agitated reactor vessel (P-17/V-107), where it is acidified with dilute sulfuric acid ("H_2SO_4 (10% w/w)") to form a precipitate of calcium sulfate (gypsum):

$$1(\text{Calcium Citrate}) + 3\,H_2SO_4 \rightarrow 2(\text{Citric Acid}) + 3\,CaSO_4 \qquad (11.3)$$

A third filter (P-18/RVF-103) is used to remove the precipitated gypsum and yields an impure citric acid solution. Careful control of pH and temperature of the precipitation steps is important for maximizing the yield of citric acid. The resulting solution is concentrated and crystallized using a continuous evaporator/crystallizer (P-19/CR-101). The crystals are recovered by filtration (P-20/RVF-104) and dried in a rotary dryer (P-21/RDR-101). If the final product is required in high purity (e.g., for pharmaceutical applications), treatment with activated carbon may precede crystallization in order to remove colorants. In addition, ion exchange is sometimes used to remove ionic species (Berovic & Legisa, 2007). These operations are not included in this model.

11.3 Model setup highlights

The basics of setting up a batch process model in SuperPro Designer have been explained in Chapter 10. In this section, some advanced process modeling features are discussed.

11.3.1 Material charges

Modeling of material charge operations is done with the help of a Charge (for raw materials) or Transfer-In (for materials fed from another procedure) operations. However, if the required amount of fed material is not known a priori, a "Pull-In" operation can be used instead. Pull-In operations calculate the material amount based on some user-defined specification. For instance, Fig. 11.2 shows how the second water charge operation is set-up in P-1/V-101 in order to dilute the glucose contents to 20%. Other specification objectives that could be used in a Pull-In operation include the specification of target feed/contents ratio, target final amount, target temperature, etc. (see Fig. 11.2).

Similar setup is necessary for the fermenter P-11/FR-101 where the amount of nutrients needed is proportional to the amount of molasses fed. Therefore, nutrients need to be *pulled* in the fermenter at a required amount which must be related to the flow of molasses. This is achieved by modeling the charging of nutrients in the fermentation procedure (P-11/FR-101) as a Pull-In operation where a 0.08 nutrients/molasses ratio must be achieved. The required amount for nutrients, as calculated at solution time, is back-propagated to the mixture preparation procedure P-7a/MX-101. This is a mixing step with regulated inputs so that a certain mixture amount and composition are achieved in the

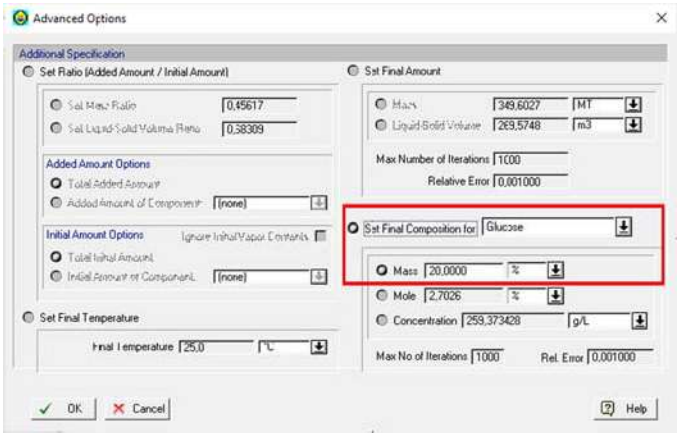

FIGURE 11.2 The advanced options dialog of a Pull-In operation for mixing unit P-1/V-101.

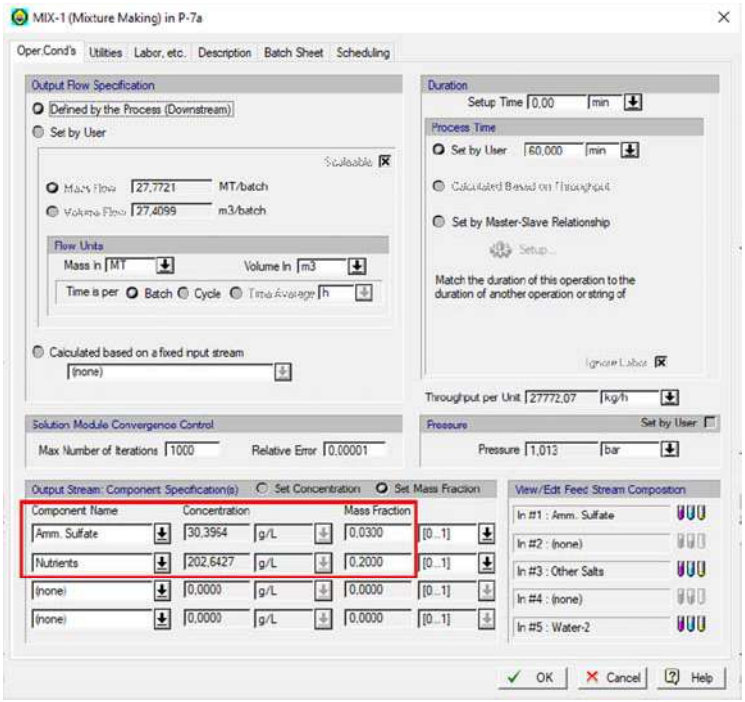

FIGURE 11.3 The operating conditions dialog of a Pull-In operation in mixing unit P-7a/MX-101.

output. As shown in Fig. 11.3, Ammonium Sulfate and Nutrients should be fed in P-7a/MX-101 so as to achieve at the output mass fractions of 0.03 and 0.2, respectively. Input streams whose amount is to not be determined at solution

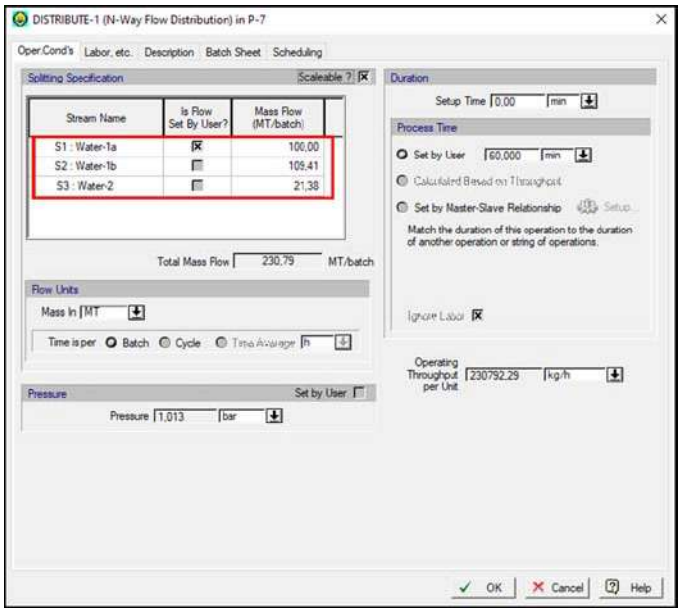

FIGURE 11.4 The operating conditions dialog of Flow Distribution operation P-7/FDIS-101.

time by the process (e.g., "Amm. Sulfate" and "Other Salts" streams in P-7a/MX-101) are labeled as "Auto-Adjust."

Water used in the plant originates from a central distribution system modeled by a flow distribution procedure (P-7/FDIS-101) which is used to feed water to the molasses tank (P-1/V-101) and the nutrients mixing process (P-7a/MX-101). In Fig. 11.4, it is shown that in the Flow Distribution operation, the amount of one output stream (Water-1a) is fixed at 100 MT/batch (the fixed amount of water fed initially into the molasses tanks), but the amounts of the other two output streams ("Water-1b" and "Water-2") are determined by the downstream process. With this set-up of pulling operations, the amounts of water, nutrients, and ammonium sulfate needed per batch are determined by the amount of molasses fed.

11.3.2 Modeling the fermentation step

The batch fermentation of molasses to citric acid is executed through a series of consecutively executed operations (Fig. 11.5): transfer-in of the molasses, pull-in of nutrients, inoculation, fermentation, transfer-out of broth, and CIP (clean-in-place) of the fermenter.

The fermentation operation is executed while maintaining a constant temperature of 35°C; since the fermentation is exothermic, cooling with Cooling Water is required to maintain the above temperature. The required

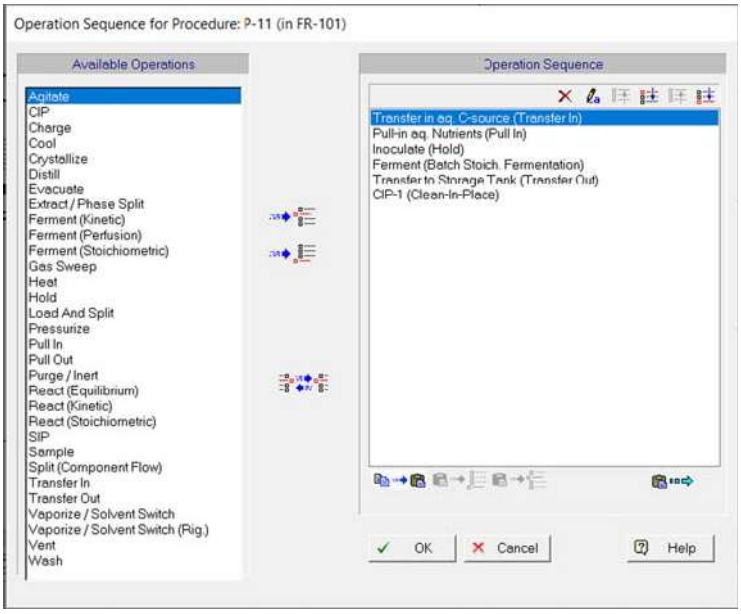

FIGURE 11.5 The declaration of operations in the Fermentation procedure P-11.

cooling duty and agent rate are calculated by the energy balance. Fermentation lasts for 5 days.

The oxygen required for fermentation is supplied by an air stream (S-114). Air is first compressed and filtered before fed into the fermenter. The rate of air needed is calculated to achieve 0.3 VVM (volume of air per volume of broth per minute). All these specifications are entered through the fermentation operating conditions dialog shown in Fig. 11.6.

Fig. 11.7 shows how the reaction stoichiometry is defined with the reactants in the left table and products in the right. The stoichiometry can be declared either on a mass or molar basis; in the latter case, proper molecular weights are required for all components participating in the reaction to check that the mass balance between reactants and products is satisfied.

In SuperPro Designer, there exist multiple options for entering a reaction scheme depending on the type of information available. In the simplest case, a "stoichiometric" operation allows the definition of (parallel or sequential) reactions using only their stoichiometry and final extent. If the kinetics of the reactions is known, then a "kinetic" operation can be used. In this case, the kinetic expression type and parameters must be set (Fig. 11.8). For fermentation kinetics, the generic reaction rate expression contains different terms (Monod, Haldane, Inhibition, etc.) which are activated and their parameters supplied as needed. In the kinetic case, the reaction extent is predicted by the model by solving the differential conservation equations over time.

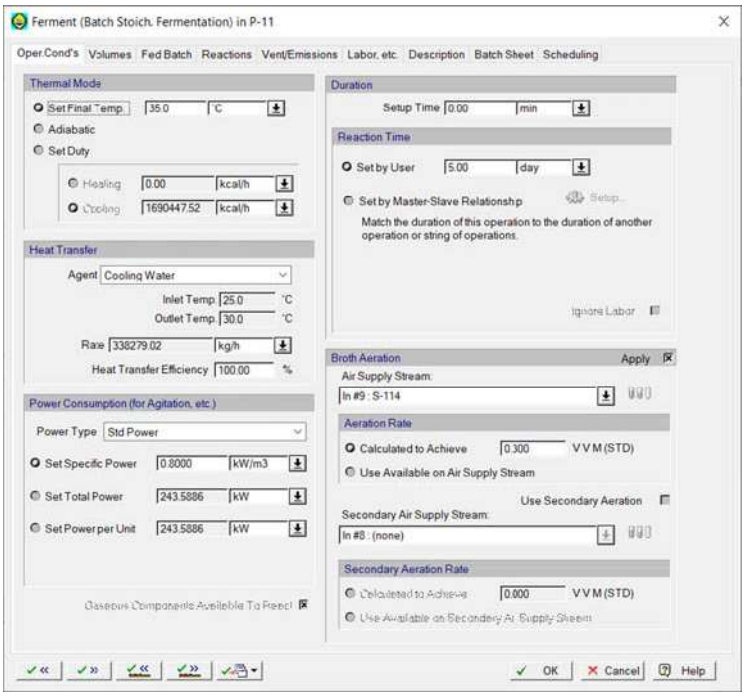

FIGURE 11.6 The operating conditions dialog of the Fermentation operation in P-11.

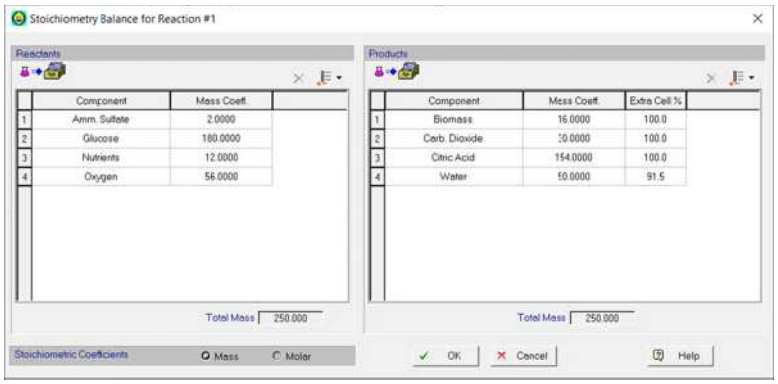

FIGURE 11.7 The fermentation reaction stoichiometry dialog for the Fermentation operation in P-11.

In this case study, the "stoichiometric" approach was used; as shown in Fig. 11.9, a conversion rate of 99% with respect to the reaction limiting component (determined at solution time) was assumed. An exothermic heat of reaction of −2990 kcal/kg of citric acid at 35°C is declared.

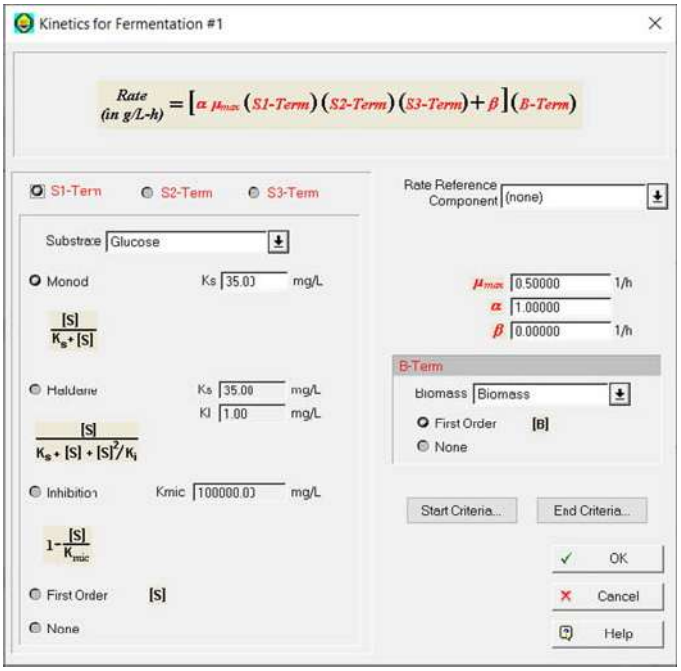

FIGURE 11.8 The kinetics dialog for a Fermentation operation.

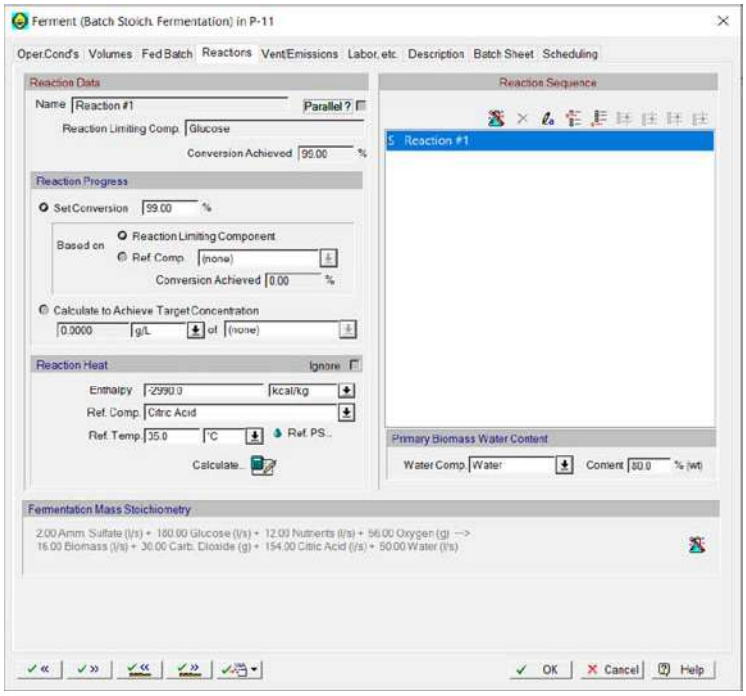

FIGURE 11.9 The stoichiometric fermentation reaction dialog for the Fermentation operation in P-11.

11.3.3 Modeling the cleaning operations

The fermentation and holding tanks need to be cleaned at the end of their use in every batch. Even though cleaning operations are not processing steps, they have to be accounted because they utilize material resources (cleaning agents) and they occupy time of the cleaned equipment and any auxiliary equipment (e.g., CIP-skids) used.

A cleaning operation may involve multiple steps. As shown in Fig. 11.10, the cleaning of fermenter includes a water flush, a caustic solution wash and a second water flush. The type and amount of cleaning agent used in each step along with its duration are declared through the dialogue in Fig. 11.10. A CIP-Skid can be also defined as auxiliary equipment used.

The waste generated by the CIP process is collected in the "CIP Waste" storage unit. Storage units are material collection points for the supply of raw material, or the deposit of product/waste streams. By specifying a replenishment/discharge strategy, it is possible to create the inventory profile of these units and use this information to properly design their capacity.

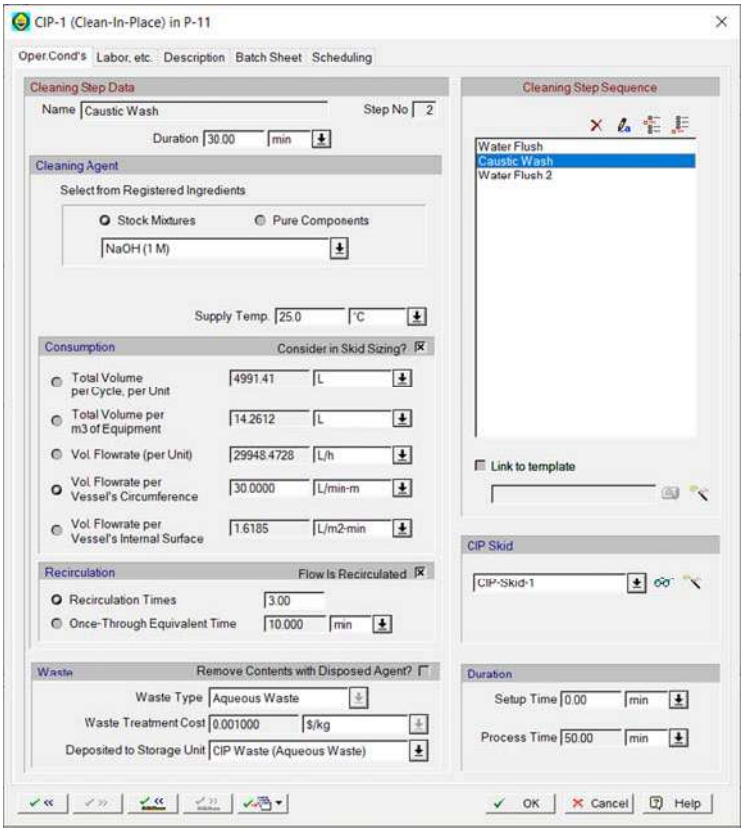

FIGURE 11.10 The CIP operation definition dialog in P-11.

11.4 Scheduling setup

As explained in Chapter 10, in a cyclically repeating batch process, the *recipe cycle time* is defined as the time period between the start of two consecutive batches. The *minimum recipe cycle time* is the cycle time threshold value below which overlaps in the utilization of equipment are unavoidable. If there is no equipment sharing among procedures of the same batch or batch integrity is maintained (i.e., equipment cannot be used by subsequent batch unless it has executed all procedures belonging to the current batch), the minimum cycle time can be calculated analytically. In all other cases, the minimum cycle time cannot be expressed by a simple equation. The subsequent sections describe how SuperPro Designer handles such complex scheduling scenarios.

11.4.1 Operating in staggered mode

Fermentations are typical examples of processes that operate in *staggered mode* (i.e., using equipment alternating among batches). Because of their long process times, they are usually the process bottlenecks. The use of multiple fermenters in staggered mode reduces the process cycle time and makes more efficient utilization of the downstream equipment.

In the citric acid case, the fermentation procedure lasts for 6.65 days including the actual fermentation and all other operations (e.g., material transfers). Fig. 11.11 shows the Equipment Data dialog for the fermentation procedure where six extra fermenters are assumed to be used in staggered mode. With a total of seven fermenters available, the effective cycle time of the fermentation procedure drops below 1 day, which makes possible the execution of a new batch every day (recipe cycle time = 1 day).

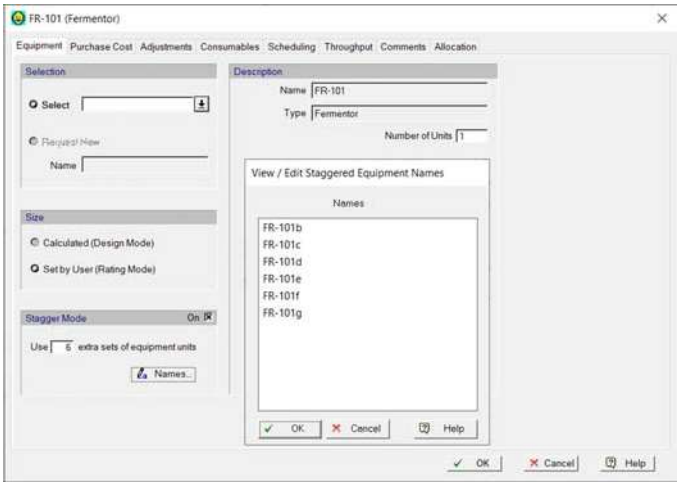

FIGURE 11.11 The equipment properties dialog of the Fermentation procedure.

It should be noted that staggering of extra equipment units is possible not only among batches but also among multiple cycles of the same procedure in the same batch (e.g., when the material to be processed cannot be handled by a single equipment unit and must be split into different "cuts"). A "carousel" of chromatography columns is a typical example of such a scenario: at every instant, one column is loaded, another is eluted, a third is regenerated etc. and the process is cyclically repeated by alternating columns. In SuperPro Designer, staggered equipment can be used to handle such cases as well.

11.4.2 Operating with independent cycling

Independently cycling procedures repeat themselves with a frequency which is different than that of the main process. This means that an independently cycling procedure processes a material amount which is different than the batch amount. If, for example, the overall process has a cycle time of 24 h and there is an independently-cycling procedure with a cycle time of 48 h, then the amount of material processed in one batch of this procedure must correspond to two batches of the main process. In this case, time is traded for capacity since the cycle time is longer but the batch size (and, consequently, the capacity of the used equipment) is larger than that of the main process.

In this case study, the vacuum filtration procedures (P-14, P-16, P-18, and P-20) are assumed to operate in independent cycling mode. In the Procedure Data dialog (Fig. 11.12), users can define a procedure's cycle time either directly or set it equal to the duration of some operation or other independently cycling procedure. Using the latter option, one can declare a set of procedures that are synchronized with the same "clock." Instead of specifying the cycle time, it is possible to define independent cycling by setting the ratio that represents how many cycles of this procedure correspond to one cycle (batch) of the main process.

It should be noted that through the same dialog, it is possible to set the number of cycles of a "normal" procedure or declare a procedure as "continuous"; in the latter case, the procedure is assumed to be operating all the time and is ignored in cycle time calculations. In this example, the downstream reactors (P-15/V-106, P-17/V-107), crystallization (P-19/CR-101), and drying procedures (P-21/RDR-101) have been defined as continuous.

11.4.3 Calculating the minimum cycle time

Using the operation scheduling data, SuperPro Designer calculates the duration of every procedure and estimates the minimum cycle time and the process bottleneck (see Fig. 11.13). As indicated earlier, if there exist procedures that share the same equipment across batches and/or sharable auxiliary equipment, the calculation of the minimum cycle time is a more involved process which is done through the Cycle Time Calculator (Fig. 11.14). Using this calculator, all

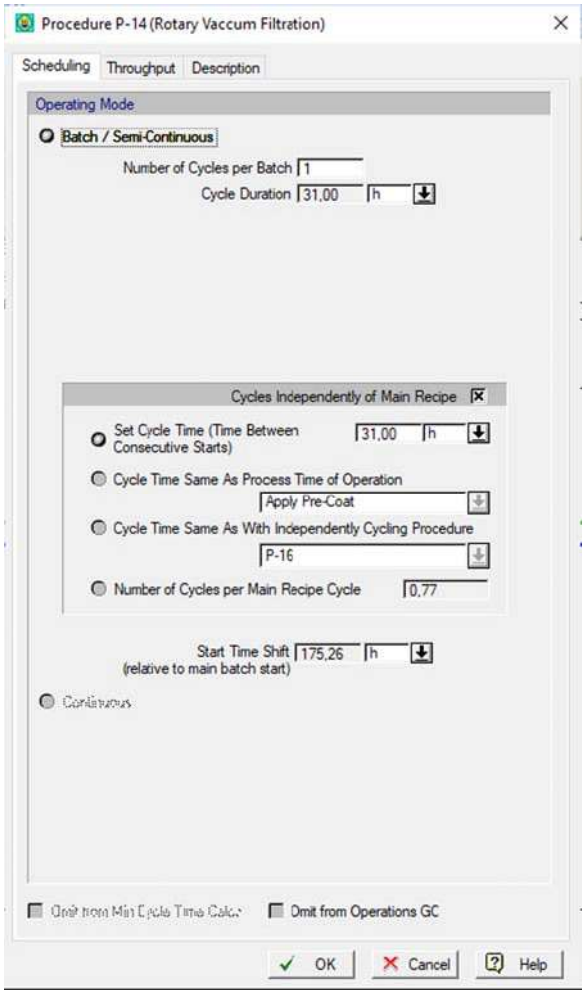

FIGURE 11.12 The procedure data dialog.

ranges of feasible cycle time values can be calculated and displayed. In this example, equipment sharing exists for the skid used for cleaning the fermenters and mixing vessels, so Fig. 11.14 shows all feasible cycle times that do not cause any skid overlap.

Users can still select a different cycle time from the calculated minimum. In this example, the minimum cycle time was calculated at 22.8 h; however, a cycle time of 24 h (a new batch every day) has been set (Fig. 11.13). This cycle time is within the feasible ranges shown in Fig. 11.14, so, no equipment utilization conflicts are expected when scheduling multiple batches of this process.

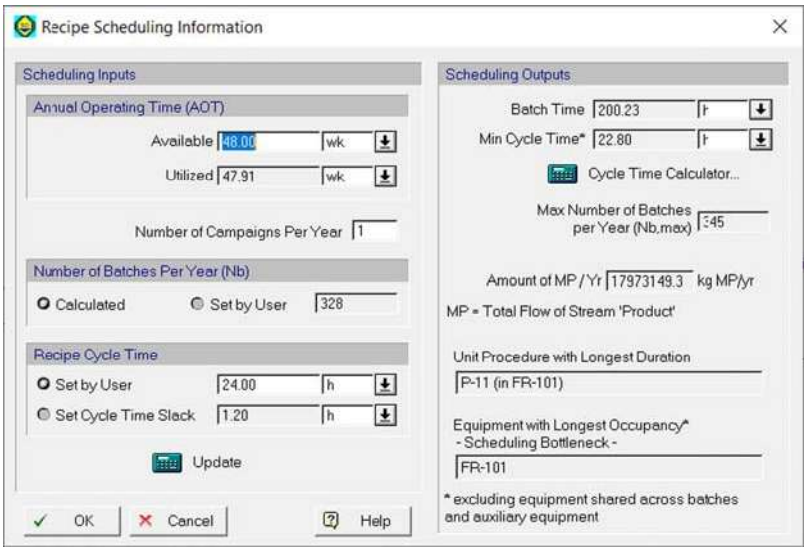

FIGURE 11.13 The scheduling information dialog and the definition of the process cycle time.

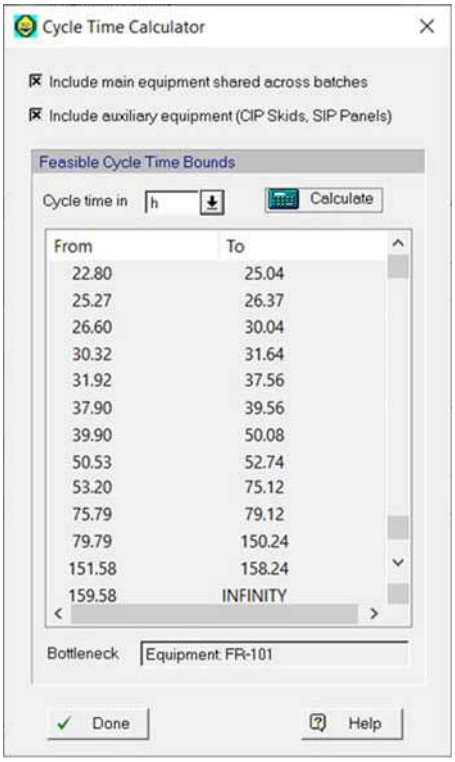

FIGURE 11.14 The process cycle time calculator.

11.5 Process simulation results

SuperPro Designer implements a sequential-modular strategy for solving the model, i.e., procedures are solved sequentially following the material flow (see Chapter 1 for details). If recycle loops exist, the tear-stream algorithm is implemented with the use of an iterative numerical solver until some convergence criteria are satisfied (see Chapter 4 for details).

The solution of balance equations for all procedures results in the calculation of the state (composition, amount, and conditions) of all intermediate/output streams and of all internal procedure states, the required amounts of utilities and the required size of all equipment.

In the citric acid case under study, for an annual production of 17,973 MT of citric acid, the calculated amounts of raw materials needed are shown in Table 11.3.

The size of all process equipment is shown in Table 11.4. Most equipment were set in design mode and their size was calculated; a few (e.g., holding tanks) were set in rating mode and their size was provided by the user. Equipment in design mode are sized to fit the requirements of the hosted operations through a "bidding" process (i.e., the most demanding operation sets the equipment size). For equipment in rating mode, SuperPro Designer checks if the provided size can satisfy the process requirements.

11.6 Process scheduling and debottlenecking

The scheduling data of every operation is the basis for determining the recipe cycle time, identifying bottleneck(s), and any potential conflict in the use of main and auxiliary equipment. The Operations Gantt chart in Fig. 11.15 shows how all operations are lined up in time for the execution of a single batch for the citric acid case.

TABLE 11.3 Raw material requirements for the entire process (MP: main product = citric acid).

Material	MT/year	MT/batch	MT/MT of MP
Water	143,024	436.049	7.96
Molasses	45,868	139.841	2.55
Amm. Sulfate	273	0.833	0.02
Nutrients	1822	5.554	0.10
NaOH (1 M)	4897	14.930	0.27
Air	273,105	832.636	15.20
Lime (33%)	35,505	108.248	1.98
H_2SO_4 (10% w/w)	149,793	456.687	8.33

TABLE 11.4 The process equipment size and cost.

Name	Type/size	Unit cost ($)
AF-101	Air filter	94,000
	Rated throughput = 27,735.4 m^3/h	
AF-102	Air filter	34,000
	Rated throughput = 10,104.8 m^3/h	
C-101	PBA column	150,000
	Column volume = 4.66 m^3	
CR-101	Crystallizer	507,000
	Vessel volume = 127.01 m^3	
FR-101	Fermenter	600,000
	Vessel volume = 350.00 m^3	
G-101	Centrifugal compressor	496,000
	Compressor power = 1468.18 kW	
PFF-101	Plate and frame filter	145,000
	Filter area = 334.72 m^2	
RDR-101	Rotary dryer	441,000
	Drying area = 82.81 m^2	
RVF-101	Rotary vacuum filter	150,000
	Filter area = 51.24 m^2	
RVF-102	Rotary vacuum filter	225,000
	Filter area = 101.53 m^2	
RVF-103	Rotary vacuum filter	194,000
	Filter area = 78.74 m^2	
RVF-104	Rotary vacuum filter	138,000
	Filter area = 41.25 m^2	
ST-101	Pasteurizer	101,000
	Rated throughput = 33,454.63 L/h	
V-101	Blending tank	340,000
	Vessel volume = 300.00 m^3	
V-102	Blending tank	340,000
	Vessel volume = 300.00 m^3	

Continued

TABLE 11.4 The process equipment size and cost.—cont'd

Name	Type/size	Unit cost ($)
V-103	Blending tank	340,000
	Vessel volume = 300.00 m^3	
V-105	Flat bottom tank	209,000
	Vessel volume = 174.39 m^3	
V-106	Neutralizer	118,000
	Vessel volume = 41.32 m^3	
V-107	Neutralizer	87,000
	Vessel volume = 14.61 m^3	

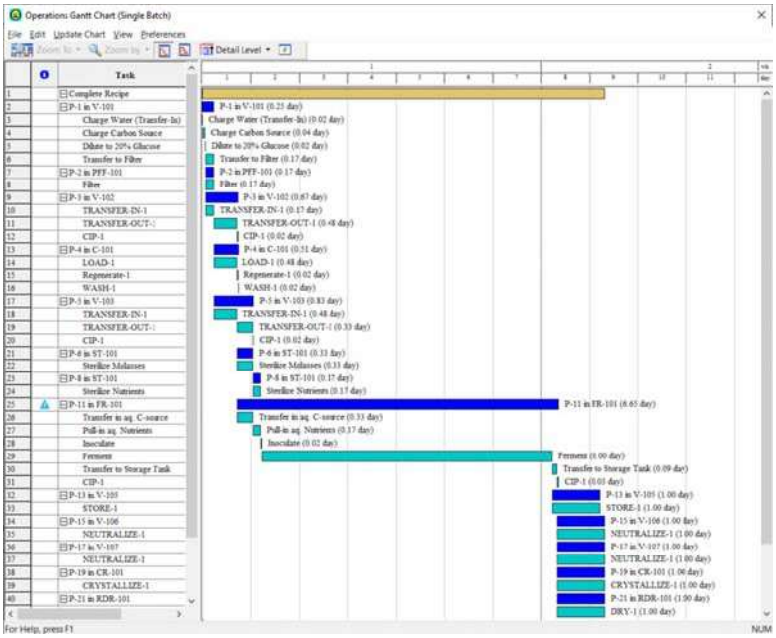

FIGURE 11.15 The operations Gantt chart.

Fig. 11.16 displays the Equipment Occupancy Gantt chart for 14 consecutive batches assuming a cycle time of 24 h; despite the long fermentation time (6.65 days), this cycle time is possible because of the use of multiple fermenters operating in staggered mode. The batch time (i.e., completion time of a batch) is approximately 8.3 days.

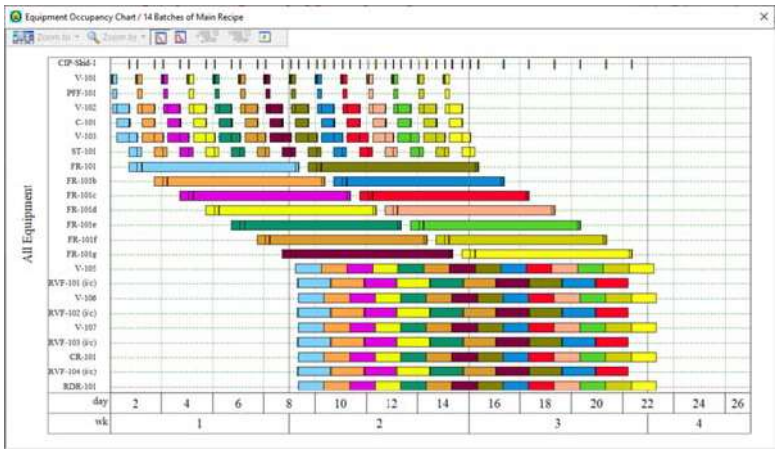

FIGURE 11.16 The equipment occupancy Gantt chart—14 batches.

Schedule-wise, the overall process is divided into the upstream which is purely in batch mode and the downstream which operates continuously. The two parts are connected through the holding tank P-13/V-105, which receives the broth from all fermenters and feeds the purification train. The absence of any gap in the utilization of the downstream equipment signifies that they operate continuously in a "back-to-back" mode. In the upstream section, fermenters are the process bottlenecks; further reduction in the cycle time is possible if more fermenters become available. In this case, extra holding tanks (e.g., V-103, V-102) and chromatography columns may also be needed (see Exercise 1). If acquiring new equipment is not possible, reduction in the cycle time may be possible by decreasing the duration of operations in bottleneck equipment by proper modifications. For example, technological advancements in increasing the fermentation efficiency and reducing the fermentation time could have significant impact on the cycle time and plant throughput.

11.7 Process economics

SuperPro Designer performs thorough economic analysis which includes the estimation of capital expenditure (CAPEX), operational expenditures (OPEX), cash flows over the project lifetime, and the values of economic evaluation indices.

11.7.1 Capital investment costs

The estimation of CAPEX is done on the basis of the total equipment cost using user-provided multiplication factors for all other costs such as piping, instrumentation, and engineering. SuperPro Designer provides meaningful

default values for all these multipliers; however, the actual values may vary considerably with the type of the plant (please refer to the SuperPro User Guide for more details).

The estimated equipment cost is shown in Table 11.4; the total equipment cost is calculated at around $9.5 million which translates to a direct fixed capital (DFC) of around $39.2 million. The total capital investment (including working capital and start-up costs) is estimated as $42.8 million.

11.7.2 Operating costs

OPEX is divided into variable and fixed costs. Variable costs are directly dependent upon the level of production and include the cost of raw materials, utilities, consumables, labor, and waste treatment or disposal. Fixed costs are facility-dependent invariable costs such as depreciation, maintenance, and insurance.

To calculate revenues, material, and waste treatment costs, SuperPro Designer needs to know the economic role all input and output streams. Through the dialog of Fig. 11.17, users can classify input streams as "raw material" or "revenue" (in case plant revenues originate from the treatment of input streams) and output streams as "revenue" or "waste" (organic, aqueous, sold, or emissions). The selling price of revenue streams or the disposal cost of waste streams can also be set in this dialogue. In this study, a selling price of $2/kg for crystal citric acid was assumed along with a disposal cost of $1/m^3 and $50/MT for liquid and solid (gypsum) waste streams, respectively.

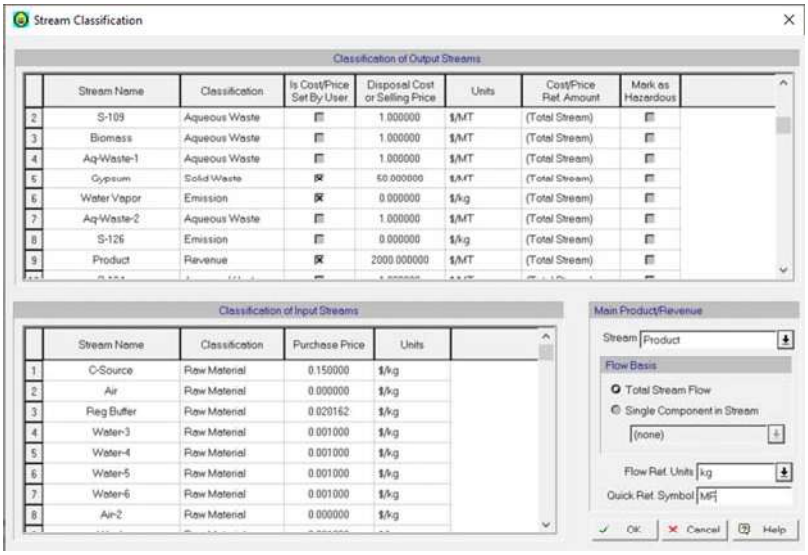

FIGURE 11.17 The stream classification dialogue for citric acid case study.

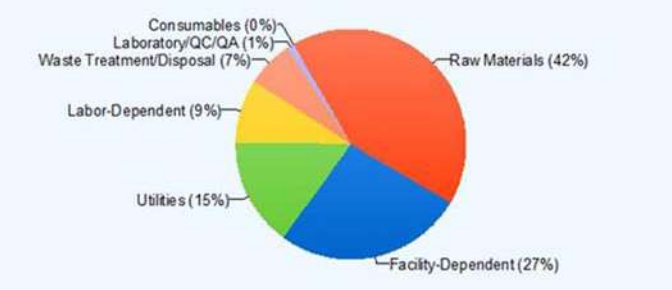

FIGURE 11.18 The annual operating cost breakdown.

Utility (heating, cooling, power) and consumable requirements are calculated from the process balances; labor requirements can be set on a per-operation or lumped-process basis. The annual costs of all these components of OPEX are calculated based on user-provided values of unit costs (e.g., per MT of steam, per labor-hour, etc.). Laboratory/QC/QA, transportation, R&D, and other costs can optionally be included.

In this case study, the total annual operating cost is estimated as $24.7 million. Fig. 11.18 shows the breakdown on all cost components in the form of a pie chart (generated within the SuperPro Designer reports). The cost of raw materials is the greatest component (42%) followed by the facility-dependent fixed costs (27%). The cost of molasses ($0.15/kg) accounts for approximately 67% of the total materials cost. The utility cost is also significant (15%). The disposal of gypsum accounts for 85% of the total environmental cost.

11.7.3 Economic evaluation

With the use of the cost and revenue values, SuperPro Designer calculates and reports various economic evaluation indices, i.e., gross margin, return on investment (ROI), payback time (PBT), internal rate of return (IRR), and net present value (NPV).

Fig. 11.19 shows all the key economic figures for the citric acid case study. The unit production cost is approximately $1.4/kg. For a selling price of $2/kg, the economic potential of the investment seems promising, with ROI at 21.6%, PBT at 4.6 years, IRR at 18.2%, and NPV 7% at approximately $41.3 million.

The effort to improve the process in order to safeguard against economic uncertainties could take different directions. One approach would be to investigate different technologies that may be more efficient and economical. As indicated before, the formation of gypsum is both environmentally and economically undesirable. Alternative flowsheets could be generated encompassing technologies such as extraction or electrodialysis membranes. In the context of process design, such alternative flowsheets could be evaluated and compared from a technological economic and environmental point of view.

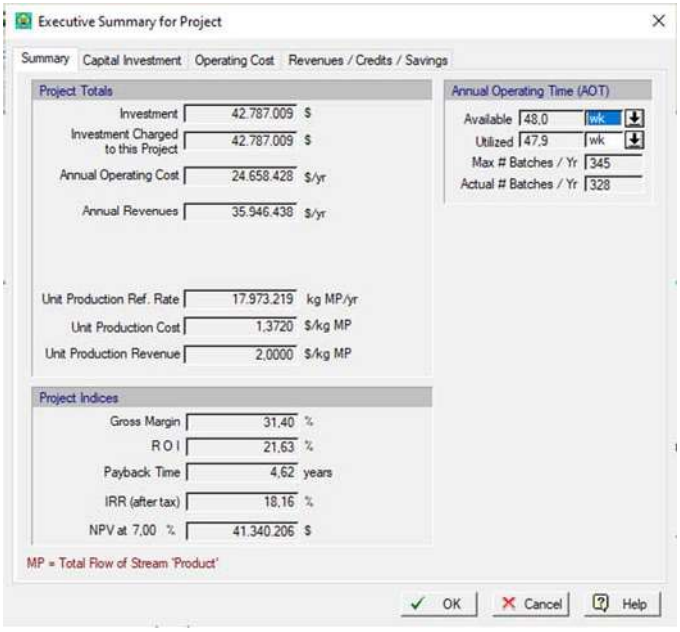

FIGURE 11.19 The executive summary dialogue with the key economic figures.

Another approach would be to seek ways to reduce operating costs, which may be done through acquiring used equipment and, thus, reducing the depreciation cost. Finding a source of cheap molasses and locating the plant near that source can reduce the raw materials cost. These types of what-if analyses are easy to perform when a simulation model of the integrated process is available.

11.8 Variability analysis

It is important to assess the effect of process variabilities on key operating, economic, and environmental variables. In SuperPro Designer, it is possible to run variability analyses with external tools with Application Programming Interface (API) functionality. Using this functionality, it is possible to change the values of input parameters, solve the balance equations and retrieve the values of output parameters through an external application while using SuperPro Designer in the background. This set-up can be used to perform sensitivity or variability analyses.

A typical way to exploit this functionality in variability analysis is to use Microsoft Excel and some add-in utilities that perform Monte-Carlo simulations such as Crystal Ball (Oracle Crystal Ball, 2021). In this case, the probability distributions of uncertain variables are provided and the Monte-Carlo simulator produces sets of random values following the provided

distribution. For each set of values, the SuperPro Designer model is solved, the results are read back and the distribution of the uncertain output variables is estimated. These analyses can be used to assess the degree to which a production system is capable of achieving specific production goals in the face of uncertainty.

To illustrate this approach, a variability analysis is executed on the citric acid case study with the objective being to identify the probability of achieving favorable economic results under uncertainty in the cost of molasses, the selling price of citric acid and the batch size. These three input variables were selected because of their significant impact on the economic bottom line and because their values are expected to show fluctuations (e.g., because of variability in the market demand and in the availability of molasses). For demonstration purpose, it is assumed that all uncertain input variables follow a triangular distribution. As shown in Fig. 11.20, the probability distribution is assumed symmetrical for batch size (within the range of 100–180 MT/batch), and skewed for the other two variables ($0.10–$0.30/kg for the molasses cost and $1.2–$2.2/kg for the product selling price). The charts in Fig. 11.20 show the assumed theoretical distribution of each variable (continuous line) and the actual distribution over 2000 samples created by the Monte-Carlo simulation.

The estimated output variables are the annual throughput (MT/yr of citric acid produced), unit production cost ($/kg of product), ROI and NPV at 7% interest rate. In the charts of Fig. 11.21, the range of expected values and the most probable values of the output variables can be extracted. Cumulative

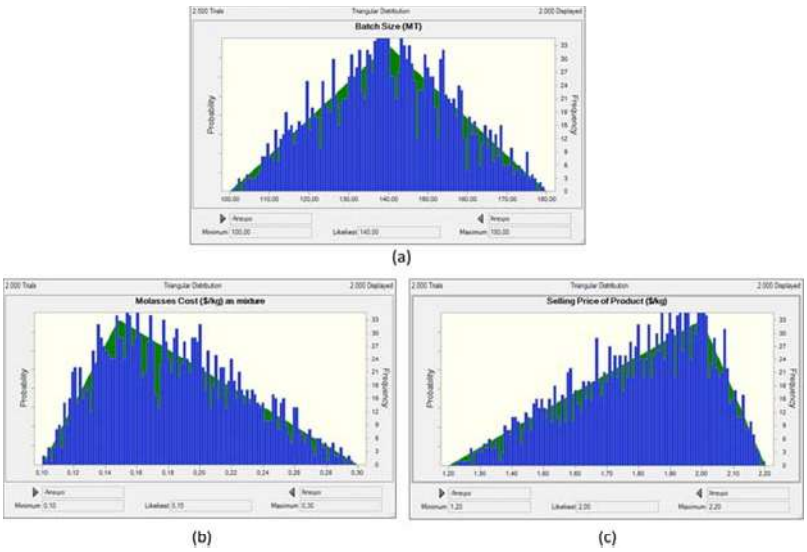

FIGURE 11.20 Probability distributions of input variables: (A) batch size (MT), (B) cost of molasses ($/kg), and (C) citric acid selling price ($/kg).

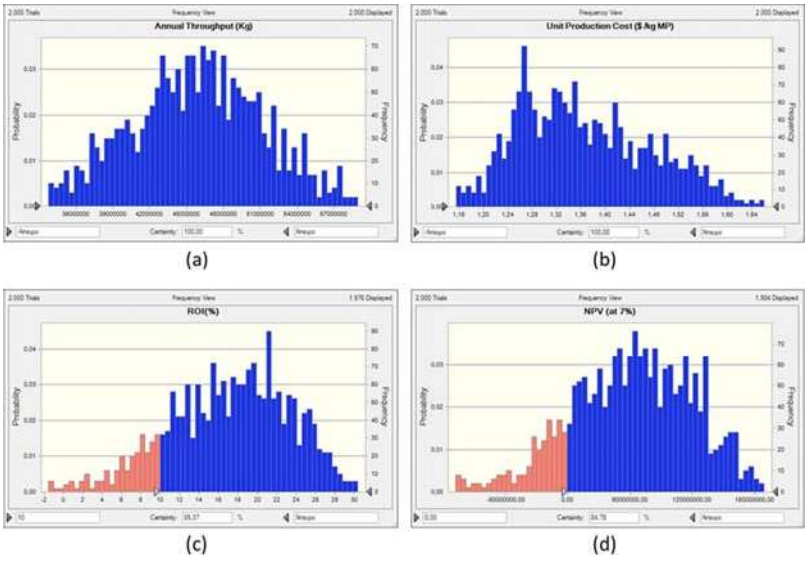

FIGURE 11.21 Predicted probability distributions of output variables: (A) annual throughput (kg/year), (B) unit production cost ($/kg of citric acid), (C) ROI, and (D) NPV at 7% interest rate ($).

probability values represent the certainty by which some target value can be achieved. For example, Fig. 11.21D shows that a positive NPV value (at interest rate of 7%) can be achieved with a certainty of about 85%. The same certainty is reported for ROI for a cut-off value of 10% (Fig. 11.21C). This threshold value, however, is very conservative in an economic viability assessment. If a higher value of ROI at 20% is sought, the certainty of achieving it in the face of uncertainty drops to 35%, indicating that high returns are not likely to be achieved with the current design. Therefore, process modifications may be needed in order to optimize the process both technologically and economically.

11.9 Conclusions

Setting up a process model in SuperPro Designer requires the declaration of all unit procedures and operations in the model, specification of their operating conditions and scheduling-related data. This chapter demonstrated how to model special operations (such are reactions, CIP, material pull-ins, flow distribution etc.) and how to set-up exceptional scheduling scenarios (such as procedures with independent cycling). It was shown how the process scheduling information can be used to estimate the process cycle time. With the latter, the annual throughput can be calculated. Process bottleneck can then be identified, which allows the reduction of cycle time to increase throughput.

Finally, it was demonstrated how modeling and scheduling results can be used as the basis for a thorough economic analysis within SuperPro Designer. With the help of an external Monte-Carlo simulation tool, variability analysis can be used to assess the effect of process uncertainties on economic and other results.

Exercises

Exercise 1: Decreasing the cycle time

As indicated in this chapter, reducing the recipe cycle time (RCT) is one way to increase the annual throughput since more batches can be executed within a year. In order to reduce the RCT, the bottleneck must be identified and eliminated with proper process modifications. In the citric acid case study, the bottleneck is the fermentation procedure; its effect on the RCT was reduced by incorporating six additional fermenters operating in staggered mode. The reduction in the minimum RCT is proportional to the number of fermenters. For example, the minimum RCT is 159.6 h (=6.65 days) when one fermenters used, and is reduced to 22.8 with 7 fermenters. It may be expected that further decrease in the RCT could be possible if the number of fermenters was increased further.

In this context, use the SuperPro Designer citric acid model, set the number of staggered fermenters to eight and re-solve the model. [NOTE: before resolving, you will need to reset the Recipe Cycle Time option from "Set by user" (24 h) to "Set Cycle Time Slack" and set this slack to 0.0. In other words, force the model to use the minimum cycle time as calculated in each case.]

a. Did the minimum RCT decrease further? If yes, was the decrease again proportional to the number of fermenters? If not, can you explain why?
b. What process/equipment modification would you propose to further decrease the RCT?
c. The above changes in the RCT are not expected to affect the equipment sizes. Yet, if you look at the estimated volume of P-15/V-106 you will notice that it increases with every reduction in the RCT. Why is that the case? (Hint: P-15 is declared as "Continuous")

Exercise 2: Increasing the batch size

Increasing the batch size may be another way to achieve higher throughput. In most cases, larger batch size means longer process times and, consequently, increased recipe cycle time (RCT). It is expected, however, that the increase in the RCT is not proportional to the increase in the batch size and the net effect on throughput will be positive. Larger batch sizes also mean increased equipment size and capital expenditures.

Use the scaling functionality of SuperPro Designer (option to "Adjust Process Throughput") and set different target batch sizes with respect to the nominal value (54.8 MT/batch of product).

a. Does the minimum RCT change when the batch size is changed? If yes, is this change proportional to the batch size change (i.e., if the batch size is doubled, is the cycle time doubling as well)? If not, why?
b. What happens to capital investment cost, operating cost per kg of final product, and the economic evaluation indices if you increase the batch size?

Acknowledgments

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